Part I. Cognitive functions and their interrelationships

Chapter 2: Sensation and Perception

In: Damasceno BP. *Research on Cognition Disorders: Theoretical and Methodological Issues* (1st Edition). Cham, Switzerland: Springer Nature Switzerland AG, 2020.

Abstract Here we analyze the first cognitive steps for understanding the world both in terms of isolated properties of an external object (sensation) or of the object as a whole (perception). The chapter presents the basic properties of sensation (modality, intensity, duration, and spatial localization) and its contributions to vital bodily functions, and discusses the question of the authenticity of sensations and perceptions. Human perception is conceived as a creative act of the individual, acquired and improved in the social and productive practice mediated by language. In perception, the object or phenomenon gets meaning by the subject´s active detection of specific and constant relationships between its different parts or properties, which requires a complex neurofunctional system constituted by various interconnected mental operations and brain regions.

Keywords: Sensation, Sensory analyzer, Perception, Gestalt Psychology, Soviet Psychology

2.1. Introduction

Sensation is a primordial act of taking cognizance of something that touches the body, is hot or cold, emit sound or light, and whose action in the sense organs is a stimulus that causes excitation of receptor cells. At the level of sensation, the living creature is already able to differentiate and generalize stimuli, notice their simple or complex character, and detect its meaning as a signal about isolated individual properties of objects and phenomena. The detection of the external thing as a whole and its meaning is a more complex cognitive process (perception), consisting in simultaneous synthesis of multiple sensory information, occurring mainly in the cerebral cortex.

2.2. Sensation

Sensation is the first step of cognition, the primary source of our knowledge about the world and its properties, movements and changes. It is the subjective image of the objective world, the effect detected, consciously or not, by the central nervous system after the psychic integration of inputs originated in the peripheral receptors and used by the organism for its orientation and activity in the external world. The psychic integration of these sensory inputs allows us to detect sounds in the air pressure waves that reach our ears, light and colors in the electromagnetic waves that stimulate the retina of our eyes, and odors in the chemical substances that excite the smell receptors of our noses. Sounds, colors, and odors are mental constructions created by sensory processing in higher divisions (cerebral cortex) of the central nervous system (CNS). Obviously, these physical and chemical excitations also arrive at lower divisions of the CNS (spinal cord, brain stem), which integrate them and generate appropriate reactions,

such as pupil constriction, eye blinking, or salivation, but we cannot be certain that we have subjective experiences of touch, light, color, smell or sound at these lower levels of the sensory pathways. The subjective experience (qualia) of sounds, odors, or colors (e.g., the redness of red roses) does not exist without the perceiving activity of a living being [1]. However, this does not mean that these subjective experiences do not authentically reflect the objective reality of the outside world, as sensed and acted upon by human beings [2].

2.2.1. The question of the authenticity of our sensations and perceptions

This has been a subject of debate and controversies between materialist and idealist philosophy. The materialist point of view is that the external world exists objectively, independent of our consciousness, with its properties exciting our receptor organs and being reflected in our mind through our sensations and perceptions. The mere existence of different types of sensations with their respective peripheral receptor organs is an ample proof that the material world exists independently from ourselves and from our consciousness. Otherwise, why and for what purpose would life and the living organisms themselves develop sensory organs, such as eyes and ears, suited for detecting even distant objects and phenomena? The more convincing explanation is that there must be things of vital importance outside, with which the organism has to interact, exchanging substances and information for survival. This fact contradicts the thesis of the subjective idealism and solipsism, which considers that only I (my consciousness) exist and that external objects are no more than sets of our sensations and ideas, only creations of our mind, without objective existence. Obviously, all things belonging to the world of culture, from artifacts and instruments to works of art, are creations of our mind, constituting a kind of objectified mind. We know the world through information from our receptor organs by processing this information in the whole aggregate of sensory analyzers mainly in the cerebral cortex, and by verifying its authenticity by means of our actions and life experience, particularly by our joint *social practice* with other human beings [2]. As asserted by Marx [3], this social practice is the criterion of truth: "The question whether objective (*gegenständliche*) truth can be attained by human thinking is not a question of theory but is a practical question. It is in practice that man must prove the truth, i.e., the reality and power, the this-worldliness (*Diesseitigkeit*) of his thinking. The dispute over the reality or non-reality of thinking which is isolated from practice is a purely scholastic question."

2.2.2. Sensations basic properties

These are (1) *modality* (vision, hearing, touch, taste, smell, and others), each with its various subtypes (e.g., sweet, salty, sour, and bitter taste); (2) *intensity*, dependent on the strength of the stimulus, whose minimal detectable intensity may vary under the influence of experience, fatigue, or context in which the stimulus is presented; (3) *duration*, dependent on the duration and intensity of the stimulus, which may change due to adaptation to a repetitive or continuous stimulus; and (4) spatial *localization*.

Each sensory *modality* is processed in its respective *sensory-perceptive analyzer*. The concept of analyzer was introduced and defined by Pavlov as “an apparatus whose purpose is to decompose the complexity of the external world into separate elements”, and “in addition to the analyzers related to external world, there must be internal analyzers, whose function is to decompose the enormous complexity of the internal phenomena arising within the organism itself, the most important of which is the motor

analyzer, the analyzer of movement…which decomposes the motor act with its enormous complexity into a large number of the most delicate elements; this ensures the enormous variety and the precision of our skeletal movements." [4].

All analyzers have a similar functional structure constituted by (1) peripheral receptors (primary sensory neurons); (2) peripheral nerves and centripetal pathways in spinal cord and brain stem conducting the peripheral excitation to the CNS; and (3) primary cortical areas for synthesis of different modalities or complex stimuli as well as efferent (centrifugal) pathways projecting from the cerebral cortex to the peripheral receptor organs. The concept of analyzer takes into account its active character in separating different characteristics of a complex stimulus or subtypes of a sensory modality, and in acting upon the peripheral receptor organ, regulating its receptive function [5]. The visual analyzer, for instance, has afferent parallel pathways originating from retinal ganglion cells and projecting to the CNS with information on form and color (in the parvocellular pathways, arisen from the small retinal ganglion cells) and on spatial relationships, depth perception (stereopsis), and motion (in the magnocellular pathways); and it has, additionally, efferent pathways projecting from the primary visual cortex to the lateral geniculate nucleus in similar way that the auditory analyzer has efferent projections from the brain stem (superior olivary complex) to the cochlea (hair cells) receptors. This conception of the visual system as a analyzer permits us to understand why only 10 to 20% of presynaptic connections of the lateral geniculate nucleus come from the retina, and the majority of its connections originate from other brain regions (feedback connections), mainly from the cerebral cortex and brain stem reticular formation [6]. The visual analyzer also includes the muscles of the eyes (the extrinsic muscles for moving the eyes in all directions and to all positions, and the intrinsic for constricting or dilating the pupils, or changing the form of the crystalline lens) as well as the proprioceptive and motor pathways for adjusting the movements and position of the eyes in such a way that the images of external objects are optimally projected onto the surface of the retina. So, the visual analyzer adjust itself to characteristics of the stimulus facilitating its reception, similarly to what occurs with the palpation movements of the hand-fingers for tactile perception of an object without the help of vision. For achieving this, the sensory-perceptive analyzer requires the participation of cortical regions involved in processes of fine tuning of the receptor organs and complex simultaneous synthesis for a more detailed and complete perception of the external object, which comprises both the object itself (its details of form and color, processed in the ventral, occipito-temporal pathways) and the context in which it is presented (spatial position and movement, in the dorsal, occipito-parietal pathways).

Sensory modality is genetically encoded in each specific receptor and its correspondent neural pathway (called *labeled line code*) in such a way that the subject always feels that sensory modality every time its receptor is stimulated, even when the stimulus is of another modality, for example, a blow to the skull or to the eye is felt as light [1].

Stimulus *intensity* and *duration* are codified by the receptor cell after its *transduction*, which consists in the conversion of the external stimulus energy (mechanical, electromagnetic) into electrochemical neural signals representing features of that stimulus. The energy of the stimulus leads to opening of ion channels (for sodium, potassium, and calcium) producing electric depolarization in a limited region of the cell membrane, with formation of a local *receptor potential* (RP). This RP is a graded response, and with increased strength of the stimulus, its amplitude becomes above the cell´s threshold for firing, and so it reaches the sensory cell´s trigger zone,

where *action potentials* (APs) are generated and transmitted through the cell´s axon to other neurons in the CNS.

Stimulus *intensity* is encoded by the frequency of discharges of APs by the sensory neuron (i.e., number of APs per unit of time, called *frequency code*). However, since the frequency of APs discharges by an isolated sensory neuron has a limit (called *saturation point*, given by the cell´s limited transduction capacity and refractory period), a stronger stimulus activates greater number of neighboring receptor cells and is then represented by a *population code*, which makes possible for the individual to sense the maximal intensity of the stimulus.

Stimulus *duration* is encoded by the patterns of discharges of APs by two types of receptor cells: the rapidly adapting (which discharge transiently, only at the beginning and ending of the stimulation) and the slowly adapting receptors (fire continuously during the whole stimulation) .

The *minimal perceptible difference* (ΔS) between two stimuli of different intensities but of same modality is determined by Weber´s law: $\Delta S = K \times S$, where ΔS is the minimal difference that can be noticed between a reference stimulus S and another stimulus, and K is a constant. As the intensity of a reference stimulus S increases, the greater the difference between it and a second stimulus must be for this difference to be noticeable, for example, we may easily detect the difference in weight between 1 kg and 2 kg but hardly between 30 and 31 kg, even though the difference remains the same (1 kg). For the intensity of a second stimulus to be detected as different, the constant K should be 1/30 of the reference stimulus S for weights, 1/10 for sounds, and 1/100 for lights [7]. The minimal detectable intensity and ΔS may vary under the influence of experience, fatigue, drowsiness, and conditions in which the stimulating object is presented [1]. Sensory-perceptive discriminative abilities are improved by social practice, particularly that of labor activities, for example, the Inuits (Eskimos) visual differentiation of many types of snow and the experienced taster´s expertise in distinguishing different qualities of wine or tea.

Localization of a stimulus (e.g., in the skin), by discriminating it from other neighboring stimuli, depends on the size of the receptors *receptive field* and on *lateral inhibition* occurring in its neural pathway. Each sensory cell has a receptive field, defined as the area of the receptor organ (e.g., skin or retina) in which the stimulation can excite the cell, leading it to produce action potentials. The size of the receptive field determines the receptor´s capacity to detect the spacing between different stimuli reaching the same receptive surface. The smaller the size of the receptive field, the greater the cell´s spatial resolution, making possible the detection of more details of the external object. For example, when reading the Braille alphabet by rubbing the index finger over the array of dots, the smaller receptive fields of Merkel skin receptors provide the most accurate distinction between the dots and the empty spaces between them. The dots cause the cell to fire action potentials, and the empty spaces silence the cell). The set of inputs from active and inactive skin receptors coming to postcentral gyrus area 3b constitutes a population code that represent a Braille letter or number.

Lateral inhibition is another neural mechanism that facilitate the precise localization of the stimulus, thus increasing the spatial resolution (Figure 2.1).

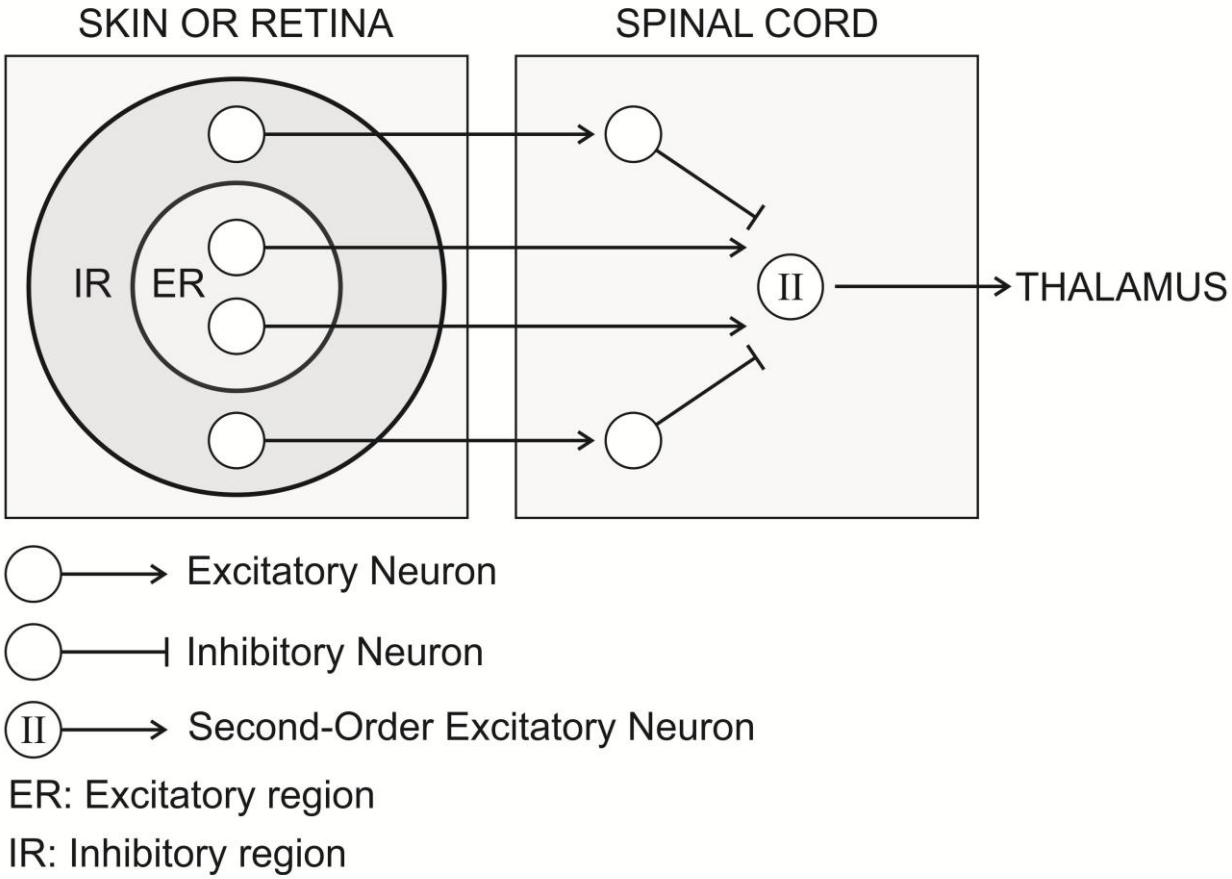


Figure 2.1. In the neural pathway conducting the stimulus excitation, the axons of various peripheral sensory cells converge onto a single second-order sensory neuron in the CNS (spinal cord or brain stem), whose receptive field is constituted by the receptive fields of all the peripheral presynaptic cells and it has a central excitatory region (ER) surrounded by a peripheral inhibitory zone (IR), which facilitate the delimitation and localization of the point with maximal stimulation.

As regards localization of sounds in the surrounding space, this is done through a neuronal population code based on the responses of ensembles of neurons which are tuned to interaural time difference of incoming signals (i.e., the interval between the arrival of the same sound at the right and left ear), becoming more accurate as the signals ascend through the auditory system, from the superior olivary complex to the inferior colliculus, auditory thalamus, and auditory cortex [8].

2.2.3. Sensory information contributes to vital bodily functions

The workings of our body, particularly of our nervous system and mind, are based on exchange of information. The term *information* is difficult to be readily defined. According to *The International Webster´s Comprehensive Dictionary of The English Language* [9], information is "any distinct signal element forming part of a message or communication, especially one assembled and made available for use by automatic

machines...". Tulving (2000) conceived it as "simply the intangible, ineffable, unknown "stuff" that is somehow created, transferred, transformed, preserved ("processed") in the mind/brain, which, when appropriately "converted", determines behavior and conscious thought" [10]. In other words, it would be any signal, sign, or meaning carrying element that is interpreted by something (machine, brain) or someone else as giving instruction for decision making or behavioral change.

Sensory information contributes to relevant bodily functions, such as homeostasis, reflex reactions, motor control, arousal, and perception. Sensations from within the body (viscera, blood vessels, skeletal muscles and joints) is used by the CNS for automatically maintaining the relative constancy of physiological processes, called *homeostasis*, such as body temperature, blood pressure, heart and respiratory rates, and blood concentration of water, sugar, salt, calcium, oxygen and hydrogen ions, etc. These metabolic processes as well as the influx of sensory excitations into the ascending reticular activating system and hypothalamus increase our state of vigilance and readiness to respond (*arousal*). Without these excitations (as in cases of long-lasting sensory isolation), the individual becomes drowsy, and then with altered state of consciousness, hallucinations, and delusions.

A new and unexpected sensory stimulus that appears suddenly in the environment, particularly if strong or biologically significant (potentially advantageous or threatening), causes orienting reactions of the organism, with turning of eyes, ears, and head in the direction of the source of the stimulus, so it can be appropriately perceived and attended to. This is the *orienting reflex*, a concept introduced by Pavlov [4].

Besides orienting reactions to stimulus first presentation, sequential stimuli or events occurring repeatedly in the environment can cause *conditioned reflexes*, which may be of two main types: the *classic* (Pavlovian), in which the animal learns to associate two or more external stimuli with each other, that is, the contingent or conditioned, and the unconditioned, innate stimulus; and the *operating* (Skinnerian), in which the animal learns, by trial and errors, to associate its own correct actions with the appearance of the innate stimulus (for instance, deployment of food when the level is pressed). The correct actions are gradually reinforced and the wrong ones, weakened (Thorndike´s "law of effect").

Sensory information, particularly visual, vestibular, proprioceptive, and tactile, is crucial for *motor control* of voluntary goal-directed movements (e.g., reaching a cup of tea with the hand) as well as gait and posture. Even the automatic movements of walking need these sensory inputs to regulate stepping.

Sensation and perception are interconnected processes, inseparable from the organisms activity, with which they constitute a functional unit. In inferior animals, sensation is enough for adaptive sensory-motor reactions to isolated properties of vitally important objects, but in superior animals living in complex and highly changeable environments, perception is necessary for representing not only the object as a whole but also the conditions in which it is located (surrounding things, obstacles, etc.).

Rubinshtein [5] pointed out that sensation, similarly to perception, is not only an image or sensitive representation, but also a component of a living being´s activity, which brings about stimuli and regulate them. Thus, the phylogenetic development of animal sensation and perception depends crucially on those biologically significant stimuli that arise in the process of the animal´s vital activity, associated to its way to act and adapt to its environment. Dogs, for instance, are excellent in detecting the odor of organic acids (arising from living bodies and trails), but they hardly perceive aromatic odors of herbs and flowers, otherwise these aromas could impede them to perceive the

odor of the prey or trail. As stated before, the sensitive-perceptive ability of humans is additionally influenced by the social character of their practical productive activity, mediated by language. Eskimos, for instance, are able to perceive 30 different types of snow to which are given 30 different names, since this differentiation is relevant for orienting their activity (hunting, gathering) in the Arctic environment. An individual´s threshold of sensitivity as well as his/her discriminative ability vary depending on various factors such as type of professional activity, level of attention, degree of adaptation to repetitive or continuous stimuli, and influence of preceding stimulus (e.g., we feel an increase of the acid flavor after tasting sweets) [5].

2.3. Perception

As we saw, sensation detects only isolated properties of objects. Perception, on the other hand, discovers sets of relevant properties of these objects and phenomena which stimulate the sensory organs. In perception, the object is represented as a whole and understood not as a simple sum of sensations caused by it, but as a set of specific and constant relationships between each of its parts and properties, here included the relations between the subject that perceives and the thing that is perceived (as conceived by Max Wertheimer, Wolfgang Köhler, Kurt Koffka, and Kurt Lewin, founders of Gestalt Psychology). An object may vary its form when seen from different angles (from the front, back, above, bottom, and sides) or when presented in different colors or sizes. A melody can be sung, performed by a musical instrument, played in different tonalities, whistled or whispered, speeded or slowed, but the relationships between its parts will stay constant (same pleasing succession of musical sounds, same rhythm), allowing the listener to recognize its meaning and, therefore, its affective value for him.

Perception is not passive or contemplative, but active, creative, dependent on the subject´s perceptive activity. Alva Noë [11] has also emphasized this enactive view of perception: "What we perceive is determined by what we do; it is determined by what we are ready to do...We enact our perceptual experience; we act it out...Perception is not just a process in the brain...whereby the perceptual system constructs an internal representation of the world...but a kind of skillful activity on the part of the animal as a whole...Perceptual experience acquires content (meaning) thanks to the perceiver´s skillful activity."

Soviet psychologists have emphasized the relevance of practical activity for the human perception, especially the social and productive practice mediated by language: (1) first, what human subjects perceive and how they perceive it depend on what they are doing and how they are doing it, as well as on the objective, content and character of their practical activity, besides being influenced by their personal characteristics such as age, educational and cultural level, profession, and interests; (2) by establishing relationships with other persons by means of language, the subject assimilates and improves the experience accumulated by society and by his/her own past experience, and this allows him/her to perceive the stimuli acting on him/her as being determined objects of reality, which belong to certain categories, such as "trees", "houses", "tools", "persons" and their respective "functions", and not as undifferentiated and indeterminate things; (3) the authenticity of sensations and perceptions is proved by the socially shared practice of the subject with other subjects, and this practice is the criterion of the truth and of the objective character not only of perception but also of all knowledge [7].

Perception is not an isolated mental faculty or mere output of a computational process in a limited brain centre, but a complex functional system comprising various interconnected mental operations and brain regions [12]. In the visual perception of an

object (e.g., a dog), we have: (1) *analysis* of fragmentary information (on color, angles of edges, movement, etc.) carried by parallel pathways from the retina through the primary visual area in the occipital lobe, and its subsequent *synthesis* in the neighboring occipito-temporal and occipito-parietal secondary visual areas, where a temporary neuronal model (neuronal assembly) is created, which constitutes the neurophysiological basis of the object´s image (percept) in construction; (2) comparison of this percept with templates of similar percepts previously existing in the long-term memory as product of the subject´s experience, without which there is no recognition or categorization of the object; (3) active and selective search for new, task relevant information, with the purpose of testing hypothesis about the nature and meaning of the object (requiring frontal-occipital interactions); (4) selection and verification of the most probable hypothesis about the meaning of the percept, by comparing it with the most congruent templates (neuronal models) existing in long-term memory; and (5) encoding of the perceived object in the phonological (dɔg) and semantic system of language. This requires a simultaneous synthesis of all essential characteristics of the object dog, both the functional (it barks, protects the owner´s house, chases cats, etc.) and the categorical ones (living being, animal), all these characteristics being processed in several co-activated interconnected brain regions.

2.4. Conclusion

Sensation is cognition first step, by which the living organism detects isolated properties of an external object, not the object as a whole, even though it can differentiate and generalize stimuli, and notice their simple or complex character. Sensations authentically reflect the world, whose objective existence (i.e. independent of our consciousness) is given by the mere fact that the organism possesses peripheral receptor organs. Sensations main properties are *modality*, *intensity*, *duration*, and spatial *localization* of a stimulus. The smaller the size of the receptive field, the greater the cell´s spatial resolution, allowing the detection of more details of the external object. The minimal perceptible difference (ΔS) between the different intensities of two stimuli of same modality is determined by Weber´s law: $\Delta S = K \times S$, where S is a reference stimulus, and K is a constant, which is 1/30 for weights, 1/10 for sounds, and 1/100 for lights. Each sensory modality has a similar functional structure which constitutes its sensory-perceptive analyzer comprising (1) peripheral receptors, (2) afferent centripetal pathways to the spinal cord or brain stem, and (3) primary cortical areas as well as efferent centrifugal projections from the cortex to the peripheral receptor organs. Sensory information contributes to relevant bodily functions, such as homeostasis, reflex reactions, motor control of voluntary goal-directed movements, arousal (vigilance and readiness to respond), and perception. Perception is the abstract representation of the object as a whole, as a set of specific and constant relationships between its parts and properties as well as between the perceiving subject and the object. The meaning of the object is grasped by means of a complex mental-cerebral functional system comprising (1) analysis of sensory information and their synthesis in an abstract percept, (2) comparison of this percept with similar percepts memorized from the previous experience, (3) selection of the most probable hypothesis about the meaning (category) of the object, and (4) its phonological and semantic encoding. Perception is not a passive reflection (as in a mirror) but a creative process, dependent on the subject´s perceptive actions, ontogenetically constructed in the joint social-cultural practice mediated by language, which allows the object to be perceived as belonging to a certain category ("tree", "house", "dog").

2.5. References

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